

# MixCDNet: A Lightweight Change Detection Network Mixing Features Across CNN and Transformer

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**Abstract**—Convolutional neural networks (CNNs) have performed notably in change detection (CD) tasks due to their superior learning and automatic feature extraction capabilities. However, they suffer from the limited receptive field and the weak modeling of long-range dependencies. Vision transformers (ViTs) excel in modeling long-range contexts and have been recently introduced in CD. Some works have combined CNN and transformers to obtain local–global information. However, these works do not fully consider the guidance and interactions from both local features (LFs) and global features (GFs). Most importantly, most of them involve a very large number of parameters and computational costs. To address these issues, in this article, we propose a lightweight CD network that mixes features across CNN and transformer (MixCDNet). We use EfficientNet as the backbone and design a novel mixing features block (MFB). First, we employ hierarchical feature extraction blocks, where local feature blocks (LFBs) and global feature blocks (GFBs) are utilized for extracting information at different spatial resolutions. Second, we propose to exploit bidirectional interactions across LFBs and GFBs branches to provide complementary clues while capturing LFs and GFs. Moreover, a skip-connection and fusion separable self-attention layer (SFSSL) is designed to obtain GFs with low complexity. Comprehensive experiments are conducted on three high-resolution remote sensing (HRRS) images CD datasets: LEVIR-CD, WHU-CD, and CDD. The results show the effectiveness of the proposed MixCDNet in improving the performance of existing CD methods with fewer parameters (0.32 M) and lower computation costs (1.59G FLOPs).

**Index Terms**—Change detection (CD), convolutional neural networks (CNNs), global–local information, lightweight CD network, mixing features block (MFB), remote sensing.

## I. INTRODUCTION

**R**EMOTE sensing images (RSIs) change detection (CD) aims to detect and localize changes of interest in a scene based on RSIs captured at different times. The CD has been widely used in numerous applications, such as urban expansion [1], disaster assessment [2], and resource surveys [3].

With the development of satellite sensors, the availability of high-resolution remote sensing (HRRS) images has

gradually increased. The HRRS images contain rich spatial structures and texture information, allowing for a more precise representation of object sizes and shapes, and their relationships. In the CD task, the HRRS images can achieve fine-grained CD results. However, they also face challenges in the following three aspects.

- 1) Due to different acquisition times, there are typical radiometric differences between bitemporal data. These differences may result in false changes caused by shadows, cloud cover, and other factors, thereby affecting the performance of the CD.
- 2) Rich object details increase the data storage requirements, algorithm complexity, and computational load.
- 3) The high complexity of noise makes it more challenging to ensure the integrity of change areas.

Early traditional CD methods utilize handcrafted features to obtain change information, with techniques such as principal component analysis (PCA) [4], change vector analysis (CVA) [5], [6], and support vector machines (SVMs) [7]. However, the handcrafted features rely on empirical design and it is difficult to extract deep features for an accurate modeling of changes between bitemporal images. This leads to low robustness and high noise in CD maps, especially for high-resolution images. Furthermore, these methods often rely on direct comparisons, such as setting thresholds or using clustering methods to obtain the final change map. This may lead to the accumulation of errors in the final results.

Different from traditional CD methods, deep convolutional neural networks (CNNs) can extract deep semantic information, providing enhanced noise resistance. Currently, most methods [8], [9], [10], [11] adopt a fully convolutional network (FCN) for end-to-end learning, allowing inputs of arbitrary image sizes. To capture change information, correlating information present in bitemporal images is crucial. Some methods [8], [9], [12] directly concatenate bitemporal images and input them into a feature extraction network. However, these approaches may suppress the unique features of different temporal phases. Most methods [8], [10], [13], [14], [15], [16], [17], [18] use Siamese neural networks, initially giving bitemporal images as inputs to two branches and finally obtaining multiscale features. The features from the two branches are usually associated through either concatenation [8], [10], [13], or differencing [8], [11], [14]. The features across different scales are commonly integrated through skip

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connection [15], [16], dense connection [13], [17], dense skip connections [18], etc.

Nevertheless, purely CNN-based models are constrained by the reception field size. To alleviate this limitation, some researchers attempt to enlarge the receptive field using dilated convolution and various attention mechanisms. In [19] and [20], dilated convolution is applied to extract dense features for CD. In [21], [22], and [23], spatial and channel attention are utilized to re-weight the fused features to model long-range relationships. Additionally, spatial-spectral [24] and spatial-channel [25], [26] cooperative dual attention modules are introduced to extract richer features. However, the methods that simply use spatial attention, channel attention, and their variants are still far from modeling global dependencies.

The transformer models have been adopted for capturing global dependencies. However, often they overlook the specific local characteristics of images due to the fact that images are split into patches in the transformer-based models. Therefore some hybrid methods [27], [28], [29], [30], combined CNNs and transformers have been proposed to capture both local and global information, achieving excellent performance. Nevertheless, these hybrid methods lack a direct mixing of local details and global semantics from CNNs and transformers. Common approaches involve simple cascading, parallel processing, and series-parallel architectures for integrating features from CNN and transformers, resulting in less interaction between these two types of features. In addition, these hybrid methods are often accompanied by a large number of parameters and high-computational complexity. This is the case of the CNN-transformer-based method ICIF-Net [29] that requires about 23.84M parameters (Params.) and 25.41G floating-point operations per second (FLOPs) to process bitemporal images of size  $3 \times 256 \times 256$ . The large number of parameters and computational demand are not suitable for real-world RSI processing, due to the limited energy and computing capacity on drones or satellites and the limited number of training samples.

To address the above issues, we propose a lightweight hybrid network, by designing a new way to efficiently mix features across CNN and transformer to obtain richer local-global information. The proposed MixCDNet initially extracts hierarchical local features (LFs) and global features (GFs) from CNNs and transformers. Then, the spatial interaction supplies local details to GFs while the channel interaction provides global semantics to LFs. The interactions provide complementary clues for a better representation of learning in both local and global blocks. Besides, the above designs are light-weighted. The proposed MixCDNet achieves a lightweight solution in the following three aspects: i) it employs the lightweight backbone network EfficientNet to extract multiscale features; ii) it designs a novel feature fusion method that allows an efficient interaction between LFs and GFs; and iii) MixCDNet adopts separable self-attention with two separate linear computations instead of the quadratic multiheaded self-attention (MHA) to encode global information.

The main contributions of our work are summarized as follows.

- 1) We propose a lightweight CNN-transformer hybrid network named MixCDNet for RSIs CD, which is characterized by only 0.32M parameters and requires 1.59G FLOPs. MixCDNet employs both a lightweight backbone and hierarchical feature extraction blocks to extract LFs and GFs.
- 2) We design an effective and efficient bidirectional interaction mechanism to mix features across CNN and transformer, enhancing the model's ability to capture local-global information by providing complementary clues for both LFs and GFs.
- 3) A skip-connection and fusion separable self-attention layer (SFSSL) is introduced for capturing GFs with low complexity.

The rest of this article is organized as follows. Section II reviews the related work about CNN-based methods, CNN-transformer hybrid networks and lightweight networks in RSIs CD. Section III introduces the proposed method in detail. Section IV presents the experimental results and discussions. Finally, conclusions are drawn in Section V.

## II. RELATED WORK

### A. CNN-Based RSIs CD Methods

In recent years, CNNs have shown good performance in CD tasks due to their powerful feature extraction capability. Three end-to-end architectures based on the U-Net model were first proposed [8]. Since then, many CNN-based methods have been developed. To further improve the feature extraction of CNN-based methods, some methods focus on extracting multiscale spatial information [31], [32], [33], [34], while others focus on capturing temporal information [35], [36], [37], or integrating features from different stages [38], [39]. MACGGNet [32] utilizes four convolutional layers with different size convolution kernels to cover ground targets with different shapes and sizes. UNet-LSTM [35] introduces long short-term memory (LSTM) into the U-Net architecture to learn the temporal correlation. FCD-R2U-Net [38] uses residual and recurrent units for feature accumulation. TFI-GR [39] integrates multilevel features by a series of guided refinement modules to capture rich context information.

Despite the impressive representational capability of CNN-based methods, they emphasize local modeling, often facing challenges in effectively representing explicit global relationships due to the intrinsic locality of convolution operations. The extraction of global information is very important in the field of remote sensing because it can provide the overall scene context information of RSIs and can better distinguish changes of interest from illumination changes, seasonal changes, or other interference sources. To achieve better long-range context modeling, some techniques have been introduced to CNN-based methods, such as in [19], [20], [21], [22], [23], and [24]. MSPSNet [19] designs a parallel convolution structure with different dilation parameters to enlarge the receptive fields. DSAMNet [21] integrates convolutional block attention modules (CBAM), and DDCNN [22] constructs a dense attention mechanism, both aim to enhance long-range information transmission. DESSN [24] proposes a

spatial-spectral nonlocal module to improve long-range dependencies. However, the above methods that utilize dilated convolution operation or integrate attention modules to the CNN-inspired architectures still struggle to obtain global information.

### B. CNN-Transformer Hybrid Networks

Transformer is originally applied to solve seq-to-seq natural language processing (NLP) problems by modeling global dependencies [40], [41]. Inspired by the excellent performance in NLP, vision transformer (ViT) [42] has been proposed to apply to computer vision tasks such as classification, segmentation, and object detection, owing to their powerful GF extraction capability. Compared to NLP tasks, the input of visual tasks is more complex due to its inherent 2-D structure. Usually, images are split into patches in transformer-based models. However, during the process of patching the input image, the 2-D neighborhood structure information within images gets disrupted. Therefore, the transformer easily ignores the local characteristics of the image. Some excellent methods have been proposed to mitigate this issue, such as the Swin transformer, and CNN-transformer hybrid structure.

SwinSUNet [43] extracts long-term global information in space-time based on the pure transformer, which uses the Swin transformer to calculate self-attention in a local window. BIT [27] firstly uses CNN to obtain high-level semantic features, next expresses the feature into a few tokens. Finally, it inputs these tokens into a transformer to model long-range contexts within the spatial-temporal domain. MCAN [28] extracts hierarchical features by CNN-based network, and constructs a transformer-based architecture to aggregate context information. ICIFNet [29] integrates CNN and transformer in a parallelized manner, and designs the intra-scale cross-interaction module and the inter-scale feature fusion module to communicate information between CNN and transformer. TransUNetCD [44] constructs a hybrid transformer model, which cascades the features from the transformer and CNN to learn local-global semantic features. ACAHNet [30] combines the global information from the transformer with the LFs from the CNN in a series-parallel way, and designs an AMCA module to reduce the computational complexity of the transformer.

Although the above models have achieved good performance, they are computationally intensive. SwinSUNet [43] uses multiple Swin transformer blocks, resulting in a high-computational load. ICIFNet [29] introduces the CNN and transformer backbones in two individual branches, causing massive resource consumption. In ACAHNet [30], the lightest model still has 11.3M parameters and 2.98G FLOPs. In addition, most existing CNN-transformer hybrid methods lack direct mixing features from CNN and transformer. BIT [27], MCAN [28], and TransUNetCD [44] adopt a sequential approach, feeding features obtained from the CNN into the transformer structure, which ensures efficiency but poorly considers the direct interaction of information between CNN and transformer. Hereby, this article aims to design a

more lightweight model and to enable a more direct interaction between local and global information.

### C. Lightweight Networks in RSIs CD

Most of the current networks designed for RSI CD are relatively complex, such as CNN-based architectures, especially CNN-transformer hybrid network structures. However, complex networks are not easy to use in practical applications, especially when deployed on mobile devices or embedded systems, or when in the presence of insufficient training samples.

Recently, a significant number of lightweight networks have been designed for tasks in NLP, object detection, and semantic segmentation. However, to the best of our knowledge, only a few lightweight methods have been proposed for the RSIs CD task. TinyCD [45] is a CNN-based Siamese U-Net-like model that utilizes low-level features to compare images, significantly limiting the number of model parameters. LSNET [46] constructs an extremely lightweight Siamese network as a backbone and designs a lightweight differential feature pyramid network to fuse features from the LightSiamese backbone. LightCDNet [47] exploits an early fusion approach as the backbone network with a small number of parameters. Specifically, it introduces a deep supervised fusion module to fuse dual-temporal images. A2Net [48] adopts a lightweight network MobileNetV2 as a backbone and guides the features extracted from the backbone network by progressive feature aggregation and supervised attention to identify changes. USSFC-Net [49] uses multiscale decoupled convolution to separate the convolution into a combination of spatial and channel correlations. This decoupling operation reduces the computational costs and parameter redundancy. Besides, USSFC-Net [49] utilizes a spatial-spectral feature cooperation strategy to achieve 3-D attention without any additional parameter compared with popular 2-D attention mechanisms.

The lightweight CD methods mentioned above all belong to CNN-based methods, lacking in the extraction of global information. Among these methods, TinyCD [45] has the smallest number of parameters and the lowest computational complexity, with 0.28M Param. and 1.45G FLOPs. However, TinyCD [45] relies only on low-level features, without considering GF relationships, which can result in a lack of scalability in complex scenarios. Here, our proposed method not only captures low-level features but also considers global information. Local and global information is extracted hierarchically, and a bidirectional interaction way is designed across local feature blocks (LFBs) and global feature blocks (GFBs).

## III. METHODOLOGY

### A. Network Architecture

In this section, we present the details of the proposed MixCDNet framework, which is depicted in Fig. 1. The current state-of-the-art lightweight method TinyCD [45] is selected as the baseline. Our method adopts the same backbone network EfficientNet as TinyCD and builds upon it by adding the extraction of global information. Furthermore, we design a novel and efficient way to obtain and mix LFs and GFs.

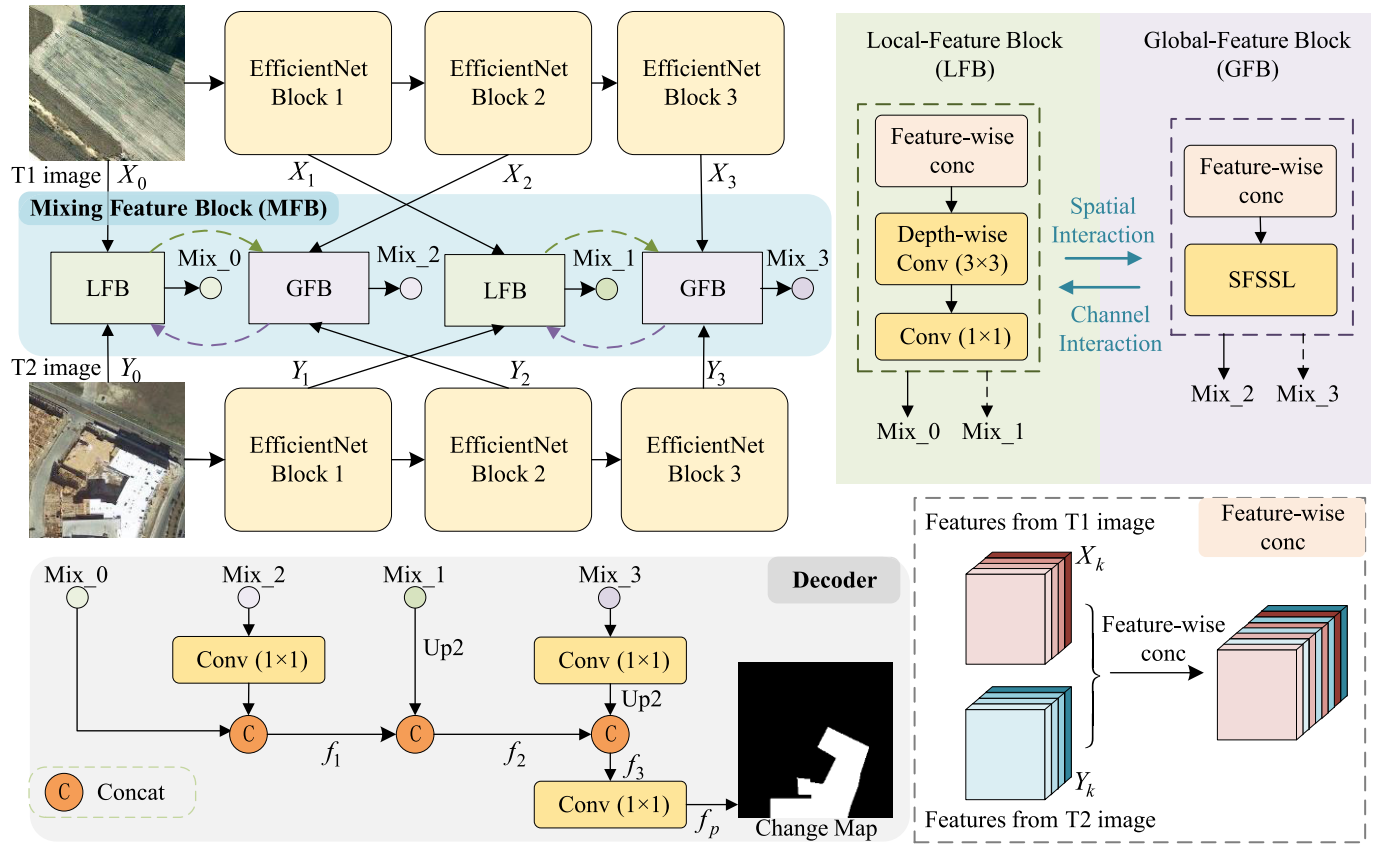


Fig. 1. Architecture of the proposed MixCDNet. The sliced version of the EfficientNet-b4 is used to extract multiscale features. Then the features of different layers of bitemporal images are fed to the LFB and GFB blocks to obtain LFs and GFs, respectively. The LFs and GFs are mixed by bidirectional interactions in MFB. Finally, the decoder uses a series of upsampling and  $1 \times 1$  convolution operations to fuse the mixed features and produce the final predictions.

As depicted in Fig. 1, MixCDNet initially utilizes the sliced version of the EfficientNet [50] as the backbone to acquire multiscale features. Subsequently, the designed LFB extracts local detailed features and generates a single-channel spatial attention map. Simultaneously, the designed GFB extracts GFs and generates a channel attention map. Notably, GFB employs separable self-attention layers (SSLs) to reduce the computational complexity. Following this, the LFs and GFs are mixed, introducing bidirectional interactions across branches (illustrated as blue arrows in Fig. 1). Finally, all the mixed features are concatenated to obtain the ultimate prediction result map.

We choose EfficientNet as the backbone due to its efficacy and efficiency. The sliced version of the EfficientNet version b4 (EfficientNet-b4) including just the first four blocks, based on ImageNet dataset [51] pretraining, are used to extract features at different stages, with downsampling rates of  $\{2; 4; 8\}$ , respectively. To preserve local detail information, the original input images and the shallow features from the first stage are used to extract local information. The last two stages are used to obtain global clues.

### B. Mixing Features Block

The proposed mixing features block (MFB) contains two key parts: 1) extraction of local and global hierarchical features with different spatial information and 2) design of bidirectional interactions across CNNs and transformers

branches. They are proposed to raise attention to change areas with different scales and strength complementary information between LFs and GFs. Details are described next.

1) *Hierarchical Features Extraction*: Firstly, we concatenate the features of different layers of bitemporal images  $X_k \in \mathbb{R}^{H_k \times W_k \times C_k}$  and  $Y_k \in \mathbb{R}^{H_k \times W_k \times C_k}$  into  $F_k \in \mathbb{R}^{H_k \times W_k \times 2C_k}$ , where  $k \in \{0, 1, 2, 3\}$ . Specifically,  $X_0$  and  $Y_0$  denote the original inputs, while  $(X_1, Y_1)$ ,  $(X_2, Y_2)$  and  $(X_3, Y_3)$  denote three different stages obtained by the sliced EfficientNet-b4. We obtain features  $F_k$  employing the following feature-wise concatenation scheme, like in [45]

$$F_k^c = \begin{cases} X_k^{(c)/2}, & c \in \text{even} \\ Y_k^{(c-1)/2}, & c \in \text{odd} \end{cases} \quad \forall c \in \{0, \dots, 2C_k - 1\} \quad (1)$$

where  $k \in \{0, 1, 2, 3\}$ .

This cascading operation enables the fusion of temporal information.

To obtain high-resolution detail information, we utilize the LFB to extract shallow features. In LFB, we only use the first two obtained cascade features  $F_0$  and  $F_1$ . We consider a depth-wise separable convolution operation (a depth-wise convolution followed by a point-wise convolution) with fewer parameters to further mix temporal and spatial information. The depth-wise convolution uses a kernel size of  $3 \times 3$  to encode local representations. The point-wise

convolution uses a  $1 \times 1$  convolution to integrate inter-channel information and to reduce the computational complexity of the model. The two stages of LF obtained by LFB can be defined as

$$\text{LF}_k = \text{Conv}_{1 \times 1}(\text{DWConv}(F_k)) \quad (2)$$

where  $k \in \{0, 1\}$  and  $\text{DWConv}(\cdot)$  denotes the depth-wise convolution.

We design the GFB to capture global context. Differently from previous global information extraction networks, e.g., [27], [28], [29], [30], [52], the proposed GFB has lower complexity. In particular, the designed SFSSL extracts GFs with linear complexity. Specific design details of the SFSSL are described in Section III-C. The obtained cascade features  $F_2$  and  $F_3$  are processed by the SFSSL module to capture two stages of GF

$$\text{GF}_k = \text{SFSSL}(F_k) \quad (3)$$

where  $k \in \{2, 3\}$  and  $\text{SFSSL}(\cdot)$  means the SFSSL layer.

2) *Bidirectional Interactions*: CNN excels at extracting local information, but it is limited by the size of the receptive field, making it challenging to obtain global information. Transformer structures, such as ViT [42], can capture the global context of input images. However, long-range attention mechanisms can easily overlook specific local characteristics of the image. Several methods have proposed various combinations of CNN and transformer architectures [27], [28], [29], [30]. However, these methods often come with high-computational complexity and a lack of bidirectional guidance between local and global information. Therefore, in this subsection, we focus on these issues.

To enhance the representational capacity of local-global information, we try to generate spatial-wise and channel-wise weights to guide GFs and LFs, respectively. Given that LFB captures LFs, it can provide complementary clues for GFB and vice versa. Thus, we propose bidirectional interactions, as shown in Figs. 1 and 2. The bidirectional interactions involve both spatial and channel interactions between the hierarchical feature extraction blocks. The LFs in the LFB flow to the GFB through the spatial interaction, which provides strong spatial clues for the GFB. At the same time, the channel interaction enables global clues to flow from the GFB to the LFB. Consequently, the proposed bidirectional interaction guides each other. In the following, the details of the bidirectional interaction design are presented.

For the spatial interaction, the goal is to provide spatial context extracted by LFB to the GFB. We utilize the pixel-wise multilayer perceptron (MLP) in [45], aiming to process all channels for each pixel simultaneously. The MLP consists of  $N$  blocks (in this article,  $N = 3$ ), and each block adopts a simple design, which consists of one  $1 \times 1$  convolution layer followed by one PReLU [53], as shown in Fig. 2. The last convolution layer only contains one filter to generate the spatial attention map with one channel. Although our spatial interaction follows the design of MLP in [45], it differs in two aspects.

- 1) *Different Application Locations*: We apply it between LFBs and GFBs, whereas the MLP in [45] is used on adjacent LFs.
- 2) *Different Objectives*: Here, we leverage the spatial attention map to compensate for the loss of spatial local details during the global information extraction process, whereas the MLP in [45] is employed to reweight each pixel to mitigate the misleading information generated by upsampling.

For the channel interaction, the goal is to provide channel context extracted by GFB to the LFB. We follow the design of the SE layer [53]. The detailed design is illustrated in Fig. 2. The channel interaction comprises a global average pooling layer (GAP), followed by two  $1 \times 1$  convolutions to change the number of channels. Between the two convolution layers, normalization (BN [54]) and activation (GELU [55]), are utilized. Finally, the sigmoid is used to generate the final channel attention. The difference between our channel interaction and SE is that the input of the attention module is different. The input of our channel interaction is derived by another branch (GFB), whereas the SE layer [53] is applied in the same branch. Same as we did in the spatial interaction, we employ channel attention to impact the local information, which provides global clues for the low-level LFs.

The spatial interaction can be formulated as follows:

$$\text{SI}_k = [\text{PReLU}(\text{Conv}_{1 \times 1}(\text{LF}_k))]_{N \times} \quad (4)$$

where  $N \times$  means that the operation is performed  $N$  times and  $k \in \{0, 1\}$ .

The channel interaction can be formulated as follows:

$$\text{CI}_k = \text{sig}(\text{Conv}_{1 \times 1}(\text{GELU}(\text{BN}(\text{Conv}_{1 \times 1}(\text{GAP}(\text{GF}_k))))) \quad (5)$$

where  $k \in \{2, 3\}$ .

The final mixed feature can be defined as

$$\begin{cases} \text{Mix\_}k = \text{LF}_k \odot \text{CI}_{(k+2)}, & k \in \{0, 1\} \\ \text{Mix\_}k = \delta_{\uparrow}^4(\text{GF}_k) \odot \text{SI}_{(k-2)}, & k \in \{2, 3\} \end{cases} \quad (6)$$

where the symbol  $\odot$  means the Hadamard product. It is worth noting that the  $\text{GF}_k$  needs to be upsampled by a factor of 4 ( $\delta_{\uparrow}^4(\cdot)$ ) so that its spatial size matches the SI map when  $k \in \{2, 3\}$ .

Specifically, the input and output sizes of spatial and channel interactions are illustrated in Fig. 2. The  $k$ th layer features from the bitemporal images are denoted by  $X_k, Y_k \in \mathbb{R}^{H_k \times W_k \times C_k}$ , where  $k \in \{0, 1, 2, 3\}$ . We assume the input size of the original bitemporal images is  $H_0 \times W_0 \times C_0$ , where  $H_0$ ,  $W_0$ , and  $C_0$  represent the height, width, and the channel numbers, respectively.

### C. Skip-Connection and Fusion SSL

Transformers are computationally intensive, primarily due to MHA, which results in  $O(k^2)$  time complexity ( $k$  means the number of tokens). The proposed SFSSL block consists of a local perception unit (LPU), a lightweight SSL, and a skip-connection and fusion (SF) block, as illustrated in Fig. 3. We will describe these three parts in the following. To reduce computational complexity, a separable self-attention approach with linear complexity [56] is introduced here.

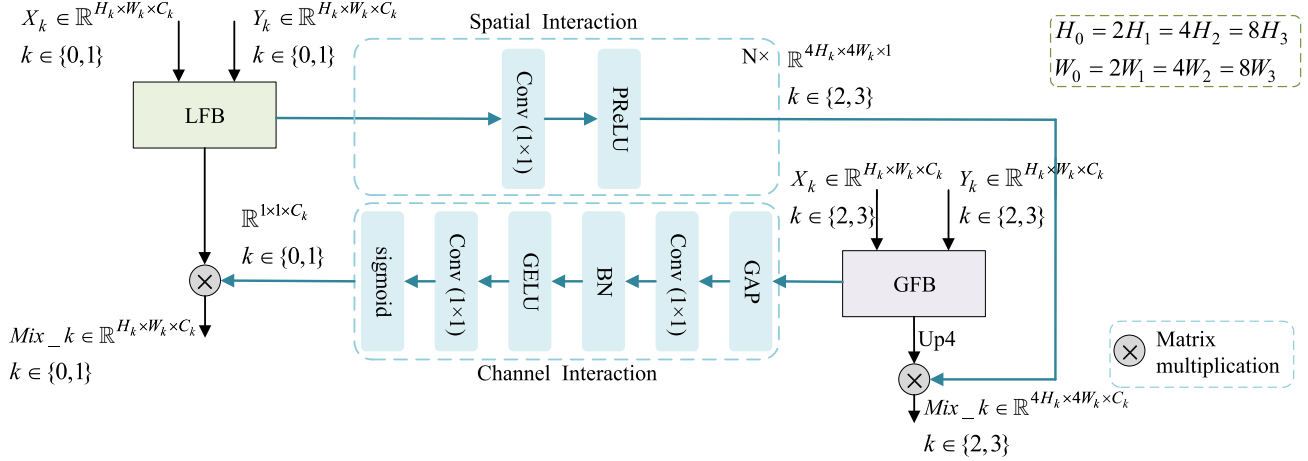


Fig. 2. Detailed design of the Bidirectional Interactions. The spatial/channel interaction provides spatial/channel context extracted by LFB/GFB to the other branch.

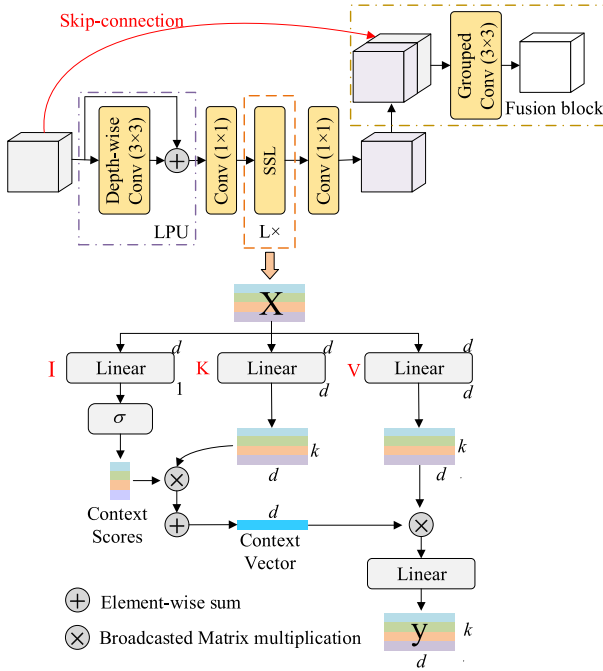


Fig. 3. Architecture of the SFSSL module. The SFSSL module consists of local representations (LPU), global representations (SSL), and a fusion block. The SSL component is linear in complexity.

1) *Local Perception Unit*: Transformer does not have the inductive bias like CNN (translation equivariance and locality between local pixels), which to some extent constrains the performance of the transformer network structure. To alleviate this limitation, an LPU is used to acquire local information, which is defined as

$$\text{LPU}(X) = \text{DWConv}(X) + X \quad (7)$$

where  $X \in \mathbb{R}^{H \times W \times C}$ .  $H$ ,  $W$ , and  $C$  represent the height, width, and channel numbers of features in the current stage.  $\text{DWConv}(\cdot)$  denotes the depth-wise convolution.

2) *Separable Self-Attention Layer*: An SSL is introduced to obtain global context with low-computational complexity. In the original self-attention module, MHA transforms an input  $X \in \mathbb{R}^{k \times d}$  into query  $Q \in \mathbb{R}^{k \times d_h}$ , key  $K \in \mathbb{R}^{k \times d_h}$  and value

$V \in \mathbb{R}^{k \times d_h}$ , where  $k$  and  $d$  mean  $k$   $d$ -dimensional token (or patch) embeddings,  $d_h = (d/h)$  is the head dimension and  $h$  is the number of attention heads. The single-head self-attention is expressed as

$$\text{Att}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{c}}\right)V \quad (8)$$

where  $\text{Att}(Q, K, V) \in \mathbb{R}^{k \times d_h}$  and  $(c)^{1/2}$  is the scaling factor to avoid gradient vanishment. Then the output of each head will be concatenated together and fed to the last linear layer  $W^O \in \mathbb{R}^{d \times d}$  to produce the multihead attention  $\text{mAtt}(X) \in \mathbb{R}^{k \times d}$ , which can be expressed as

$$\text{mAtt}(X) = (\text{Cat}(\text{Att}^1(Q, K, V), \dots, \text{Att}^h(Q, K, V)))W^O. \quad (9)$$

Similar to original MHA, the SSL transforms the input  $X \in \mathbb{R}^{k \times d}$  into three batches, i.e., input  $X_I$ , key  $X_K$ , and value  $X_V$ . The input  $X_I$  can be expressed as

$$X_I = XW^I \quad (10)$$

where  $W^I \in \mathbb{R}^d$  is a linear layer and  $X_I \in \mathbb{R}^k$  is a  $k$ -dimensional vector. Then a softmax operation is applied to the vector  $X_I$ , producing the context scores  $C_S \in \mathbb{R}^k$ .

The context scores  $C_S$  are then applied to calculate a context vector  $C_V$ . First, the input  $X$  is projected onto key  $X_K \in \mathbb{R}^{k \times d}$  by key branch  $W^K \in \mathbb{R}^{d \times d}$

$$X_K = XW^K. \quad (11)$$

Then the context vector  $C_V \in \mathbb{R}^d$  can be calculated as

$$C_V = \sum_{i=1}^k C_S X_K. \quad (12)$$

The context vector  $C_V \in \mathbb{R}^d$  corresponds to  $QK^T \in \mathbb{R}^{k \times k}$  in the above (8), yet  $C_V$  has lower computational complexity. The calculation of  $X_V$  is similar to that of  $X_K$ , the difference is that it is followed by a ReLU activation,  $X_V \in \mathbb{R}^{k \times d}$ . The context vector  $C_V$  and  $X_V$  undergo a broadcasted matrix multiplication operation, and then pass through the linear layer  $W^O \in \mathbb{R}^{d \times d}$

to obtain the final output  $y \in \mathbb{R}^{k \times d}$ . The output of SSL can be expressed as

$$y = \left( \sum_{C_V \in \mathbb{R}^d} \left( \underbrace{\text{softmax}(XW^I) * XW^K}_{C_S \in \mathbb{R}^k} * \text{ReLU}(XW^V) \right) W^O \right) \quad (13)$$

where  $*$  and  $\sum$  represent broadcastable element-wise multiplication and element-wise summation operations, respectively.

3) *SF Block*: The skip-connection operation (red arrow in Fig. 3) is employed to concatenate features from different layers. In the fusion block, a group convolution is utilized to further integrate features and reduce the number of feature channels. The skip connection, similar to the classic ResNet, serves to enhance the gradient propagation capability across layers and strengthen the representation of features. We show in ablation studies that the SF block contributes to the network achieving better results.

Please note that in Fig. 3, “ $L \times$ ” represents the number of SSLs. Here, we define  $L = 8$ . In Section IV, the selection of the  $L$  value is discussed.

#### D. Fusion and Prediction Module

The mixed features obtained at different stages contain multiscale information of the original RSIs. We aim to gradually fuse these features. The fused features are denoted as  $f_1, f_2, f_3$ , and can be expressed as

$$\begin{cases} f_1 = \text{cat}(\text{Mix\_0}, \text{Conv}_{1 \times 1}(\text{Mix\_2})) \\ f_2 = \text{cat}(f_1, \delta_{\uparrow}^2(\text{Mix\_1})) \\ f_3 = \text{cat}(f_2, \delta_{\uparrow}^2(\text{Conv}_{1 \times 1}(\text{Mix\_3}))) \end{cases} \quad (14)$$

where  $\text{cat}(\cdot, \cdot)$  denotes the cascade operation and  $\delta_{\uparrow}^2(\cdot)$  denotes an upsampling (2 times) operation.

The final pixel-level predictions, denoted as  $f_p$ , are produced by feeding the fused features  $f_3$  to a  $1 \times 1$  convolutional layer

$$f_p = \text{Conv}_{1 \times 1}(f_3). \quad (15)$$

Unchanged pixels are far more abundant than changed pixels in RSIs. To alleviate the impact of this imbalance, during the training process, we use a hybrid loss function [13], which is expressed as

$$\mathcal{L} = \mathcal{L}_{\text{wce}} + \mathcal{L}_{\text{dice}} \quad (16)$$

where  $\mathcal{L}_{\text{wce}}$  defines the cross-entropy loss and  $\mathcal{L}_{\text{dice}}$  represents the dice loss. They are expressed as

$$\mathcal{L}_{\text{wce}} = -\frac{1}{H \times W} \sum_{k=1}^{H \times W} \left( \log \left( \frac{\exp(\hat{y}[k][\text{cla}])}{\sum_{l=0}^1 \exp(\hat{y}[k][l])} \right) \right) \quad (17)$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \cdot Y \cdot \text{softmax}(\hat{Y})}{Y + \text{softmax}(\hat{Y})} \quad (18)$$

where  $\hat{Y} = \{\hat{y}_k, k = 1, 2, \dots, H \times W\}$  denotes the change map,  $\hat{y}_k$  represents a point in  $\hat{Y}$ ,  $H$  and  $W$  denote the height and width of  $\hat{Y}$ , “cla” is 0 or 1 and denotes unchanged or changed pixels and  $Y$  is the ground truth.

## IV. EXPERIMENTS

### A. Datasets

To assess the effectiveness of the proposed method, we conduct experiments on three HRRS CD datasets, i.e., LEVIR-CD, WHU-CD, and CDD datasets.

The LEVIR-CD [1] is a large-scale CD dataset, which contains 637 pairs of Google Earth images with size  $1024 \times 1024$  pixels and resolution 0.5 m. These images are taken from 20 different regions in the USA, and focus on the changes of buildings. There are many seasonality and illumination changes in LEVIR-CD. In addition, this dataset contains various types of buildings having different scales. In our experiment, we crop the images into  $256 \times 256$  pixels and adopt its default dataset split: 7120/1024/2048 for training/verification/test.

The WHU-CD [57] is a building CD dataset, which contains a pair of aerial images with a size of  $32507 \times 15354$  pixels and a resolution of 0.3 m. It covers the area in the region of Christchurch, New Zealand, which experienced a 6.3 magnitude earthquake in February 2011 and underwent reconstruction in the following years. However, there is no standardized data split solution for the WHU-CD dataset. We follow the works in [45], [58], [59], and [60]. The images have been cropped into nonoverlapping patches with a size of  $256 \times 256$  pixels: 5947/743/744 for training/verification/test.

The CDD [61] is a realistic seasonal remote sensing variation dataset that includes changes in buildings, roads, vehicles, etc., and ignores changes caused by seasonality and lighting. CDD dataset contains 11 pairs of Google Earth images with a spatial resolution between 3 and 100 cm. Following [61], the original images have been cropped and rotated to generate 16 000 pairs of images with size  $256 \times 256$  pixels: 10 000/3000/3000 for training/verification/test.

### B. Implementation Details and Evaluation Criteria

The experiments are implemented on PyTorch, using a single NVIDIA 3080 10 GB GPU for training and testing. We use the Adam optimizer with an initial learning rate of  $3 \times 10^{-3}$ , and a weight decay set to  $9 \times 10^{-3}$ . The cosine annealing strategy [62] is used to dynamically adjust the learning rate, without incorporating warm restarts. The batch size of our model is set to 8, which is limited by hardware conditions. The epoch number is set to 100. We use the Albumentations library [63] to apply data augmentation. The applied geometric augmentations include random flips along the  $X$ - and  $Y$ -axes, as well as random rotations with degrees of freedom. Additionally, nongeometric augmentations are applied including Gaussian blur and random changes in brightness/contrast.

To evaluate the performance of the proposed method, precision (Pre), recall (Rec),  $F1$ -score ( $F1$ ), overall accuracy (OA), and kappa coefficient (KC) are used. The first three

TABLE I

CD RESULTS ON THE THREE CONSIDERED TEST SETS. ALL RESULTS ARE DESCRIBED AS PERCENTAGES (%). THE BEST SCORES ARE MARKED IN BOLD. THE (+NUMBER) DENOTES THE ABSOLUTE IMPROVEMENTS RELATIVE TO THE SECOND-BEST SCORES

| Method                     | LEVIR-CD     |              |              |              |                     | WHU-CD       |              |              |              |                     | CDD          |              |              |              |                     |
|----------------------------|--------------|--------------|--------------|--------------|---------------------|--------------|--------------|--------------|--------------|---------------------|--------------|--------------|--------------|--------------|---------------------|
|                            | Pre          | Rec          | OA           | KC           | F1                  | Pre          | Rec          | OA           | KC           | F1                  | Pre          | Rec          | OA           | KC           | F1                  |
| SNUNet                     | 90.62        | 88.04        | 98.93        | 88.75        | 89.31               | 88.78        | 87.88        | 99.08        | 88.33        | 87.85               | 94.72        | 92.93        | 98.55        | 93.00        | 93.82               |
| BIT                        | 91.95        | 88.37        | 99.01        | 89.61        | 90.13               | 88.17        | 89.52        | 99.11        | 88.37        | 88.84               | 95.67        | 94.32        | 98.83        | 94.32        | 94.99               |
| ICIF-Net                   | 89.18        | 90.15        | 98.94        | 89.11        | 89.66               | 89.97        | 90.37        | 99.22        | 89.76        | 90.17               | 94.68        | 92.58        | 98.51        | 92.77        | 93.62               |
| <i>Lightweight Methods</i> |              |              |              |              |                     |              |              |              |              |                     |              |              |              |              |                     |
| LightCDNet                 | 90.83        | 90.71        | 99.06        | 89.78        | 90.77               | 91.85        | 91.01        | 99.32        | 90.68        | 91.43               | 94.23        | 91.59        | 98.27        | 91.62        | 92.89               |
| A2Net                      | 92.28        | 89.82        | 99.10        | 90.56        | 91.04               | <b>94.58</b> | 89.99        | 99.40        | 91.92        | 92.23               | 95.51        | 92.97        | 98.53        | 93.38        | 94.22               |
| USSFC-Net                  | 89.62        | <b>90.93</b> | 99.02        | 89.69        | 90.27               | 90.58        | 91.00        | 99.27        | 90.39        | 90.78               | 94.79        | 95.54        | 98.81        | 94.46        | 95.16               |
| TinyCD                     | <b>92.82</b> | 89.44        | 99.11        | 90.63        | 91.10               | 91.78        | 91.46        | 99.34        | 91.27        | 91.62               | 94.73        | 93.44        | 98.48        | 93.21        | 94.08               |
| MixCDNet (proposed)        | 92.23        | 90.78        | <b>99.14</b> | <b>91.05</b> | <b>91.50(+0.40)</b> | 94.51        | <b>92.30</b> | <b>99.48</b> | <b>93.12</b> | <b>93.39(+1.16)</b> | <b>97.52</b> | <b>96.72</b> | <b>99.26</b> | <b>96.69</b> | <b>97.12(+1.96)</b> |

indices are commonly used in CD to assess the total capabilities of CD methods. In detail, a higher precision value indicates a lower number of false alarms and a higher recall value indicates a lower number of missed alarms.  $F1$ -score balances between false alarms and missed alarms, reflecting the generalization ability of the model. They are expressed as

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (19)$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (20)$$

$$F1 = \frac{2 \times \text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \quad (21)$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (22)$$

where TP, FP, FN, and TN represent true positive, false positive, false negative, and true negative, respectively. In addition, the kappa coefficient is used to assess the agreement between the predicted result and the ground truth, expressed as

$$\text{KC} = \frac{\text{OA} - P}{1 - P} \quad (23)$$

where  $P$  represents the theoretical agreement, which can be expressed as

$$P = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN}) + (\text{TN} + \text{FP})(\text{TN} + \text{FN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2}. \quad (24)$$

It is worth noting that, besides OA and KC being utilized for evaluating the overall performance of both the changed and unchanged classes, the other metrics exclusively compute results for the changed class, as the focus of this work is primarily on the areas of change.

### C. Comparisons With SOTA Methods

To verify the effectiveness of the proposed method, seven state-of-the-art methods are selected as comparative techniques, including SNUNet [13] (which is a CNN-based method), BIT [27] and ICIF-Net [29] (which are two CNN-transformer hybrid methods), LightCDNet [47] A2Net [48], USSFC-Net [49], and TinyCD [45] (which are four lightweight methods).

**SNUNet** [13] is a densely connected Siamese network with ensemble channel attention modules, which refines features of different semantic levels to strengthen the edge of the changed target.

**BIT** [27] extracts high-level features by a CNN backbone and introduces a transformer to model global contexts within the spatial-temporal domain.

**ICIF-Net** [29] integrates CNN and transformer in a parallel manner to extract LFs and global representations.

**LightCDNet** [47] employs the early fusion method as its fundamental architecture and introduces a deep supervision-based method to fuse features from bitemporal images. We choose LightCDNet-base as the comparative method.

**A2Net** [48] extracts features by progressive feature aggregation and supervised attention with few parameters and low-computational costs.

**USSFC-Net** [49] introduces a spatial-spectral feature cooperation strategy to learn 3-D spatial-spectral attention weights without adding any parameter compared to the existing 2-D attention mechanisms.

**TinyCD** [45] uses a Siamese U-Net architecture capturing low-level feature and mixes features in the space-time domain with a low-computational complexity. As mentioned above, we use TinyCD [45] as our plain baseline.

Table I presents the results of comparison experiments on the LEVIR-CD, the WHU-CD, and the CDD datasets. As shown in the table, the proposed MixCDNet achieves the best results in terms of  $F1$ -scores. In the LEVIR-CD test set, the  $F1$ -score of our MixCDNet is 91.50%, which outperforms the second highest  $F1$  by 0.40%. In the WHU-CD test set, MixCDNet achieves values 1.16%, 1.20%, and 2.31% higher than those of A2Net in terms of  $F1$ , KC, and Rec, respectively. Moreover, as shown in Table II, the proposed method has also a relatively small number of parameters and a low-computational cost. Taking the input image of size  $3 \times 256 \times 256$  as an example, the proposed method only requires 0.32M parameter and 1.59G FLOPs, which are 11.81 times fewer and 1.92 times lower than those required by A2Net. In the CDD test set, our MixCDNet improves by 1.96%, 2.23%, and 1.18% the  $F1$ , KC, and Rec, respectively, with respect to the USSFC-Net. These comparison results prove the overall effectiveness of the proposed MixCDNet on the considered datasets.

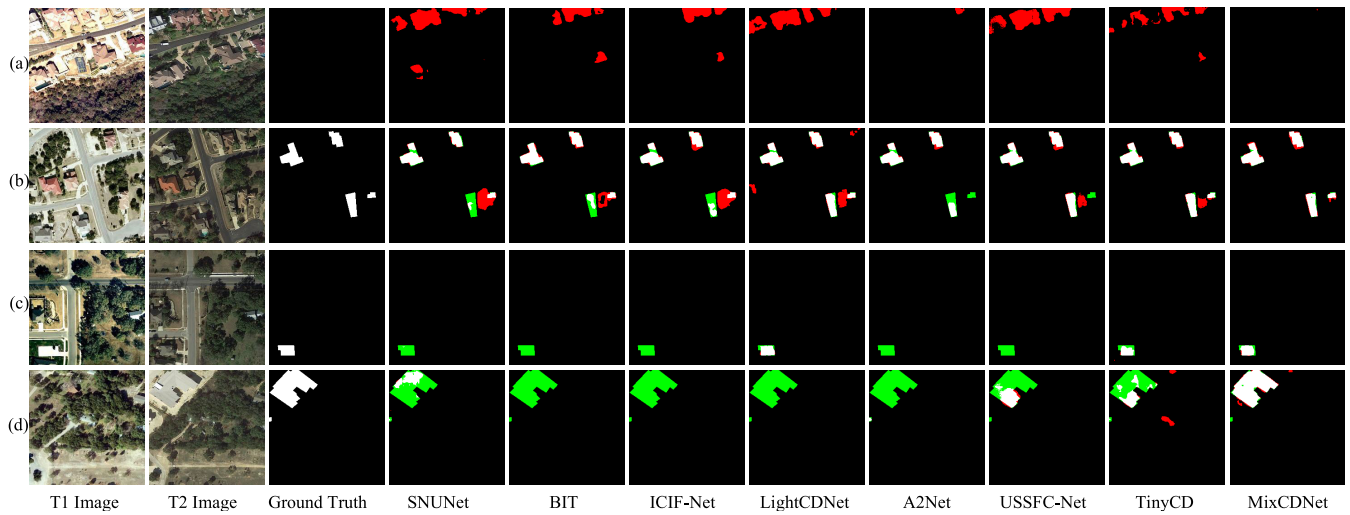


Fig. 4. Visual results of different methods on the LEVIR-CD test set. (a)–(d) Four representative samples. For a better view, white, black, red, and green represent TP, TN, FP, and FN, respectively.

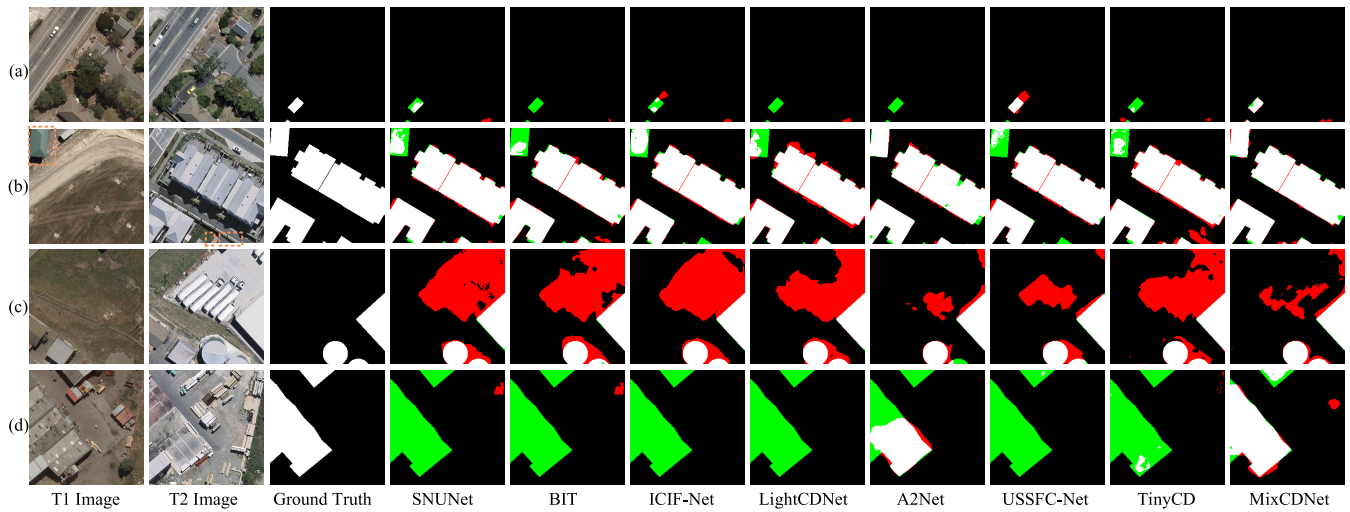


Fig. 5. Visual results of different methods on the WHU-CD test set. (a)–(d) Four representative samples. For a better view, white, black, red, and green represent TP, TN, FP, and FN, respectively.

Visual qualitative results on the LEVIR-CD are shown in Fig. 4. One can see that, in Fig. 4(a), the illumination intensity between images taken at different times is obviously different, but there are no real changes (the changes of buildings). In Fig. 4(b), there is one building that has not undergone any changes (located to the right of the changed building in the bottom area), but due to the variations in light intensity, most comparison methods incorrectly identify it as changed (view as red). The proposed method can overcome the influence of illumination, whereas other comparative methods result in a significant number of FPs. In Fig. 4(c), the changed object is small in size and has a similar appearance to the surroundings, making it less noticeable. The proposed method almost completely detects the changed target, whereas other methods generate a substantial number of FNs. Similarly, in Fig. 4(d), the newly added building exhibits a similar appearance to the surrounding background. The proposed method achieves good detection results, with fewer FPs and FNs compared to other considered methods.

Fig. 5 shows visual qualitative results on the WHU-CD. As one can perceive, in Fig. 5(a), the color of the changed building is similar to the surrounding asphalt road. This similarity results in most literature methods, such as BIT, LightCDNet, and A2Net techniques, failing to detect the changes. In contrast, our proposed method can almost entirely detect the changed object. In the locations encircled by the orange dashed boxes in Fig. 5(b) T1 and T2 images, changes have occurred in the buildings. The proposed method can identify all three changes (marked with orange dashed boxes), whereas other comparative methods either result in numerous FPs or fail to detect all three changes, especially the smallest one. There are many large cars and containers in the T2 image in Fig. 5(c). While most comparison methods mistakenly classify this area as building change (view as red), the proposed MixCDNet demonstrates its superiority in distinguishing them from buildings, thanks to the enhanced discriminant features achieved through the effective mixing of local and global information. In Fig. 5(d), the changed

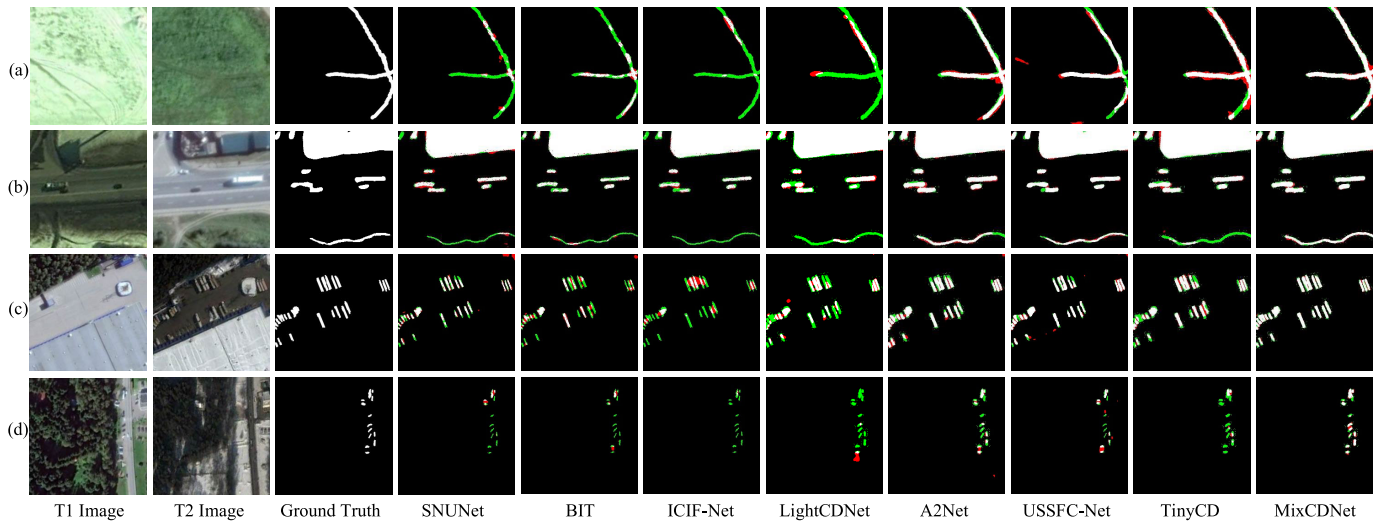


Fig. 6. Visual results of different methods on the CDD test set. (a)–(d) Four representative samples. For a better view, white, black, red, and green represent TP, TN, FP, and FN, respectively.

area corresponds to the demolished buildings, but the post-demolition regions have a similar appearance to the original building areas. These changes were not well detected by the mentioned comparative methods, but the proposed MixCDNet method can detect relatively complete change areas.

Visual qualitative results on the CDD are illustrated in Fig. 6. For images in Fig. 6(a) and (b), due to seasonal changes and variations in lighting intensity, most comparative methods struggle to detect roads comprehensively, whereas the proposed method can achieve a relatively complete detection of the road. In Fig. 6(c) and (d), which are similarly affected by seasonal changes and variations in lighting intensity, the proposed method demonstrates better detection performance for changes in dense and small-scale vehicles compared to other comparative methods. This demonstrates that the proposed MixCDNet can effectively leverage global information to capture the overall scene context and mitigate interference from illumination changes and seasonal variations. Additionally, it also indicates that the proposed method can preserve local detail features while extracting global context, enabling the detection of changes in dense and small-scale areas.

In several visual comparisons, the proposed MixCDNet demonstrates effective detection performance in complex environments, such as bitemporal images with obvious seasonal changes or significant variations in lighting conditions. Moreover, it exhibits superior performance in detecting small and densely distributed changes. This may be attributed to MixCDNet's ability to extract hierarchical LFs and GFs and effectively mutually guide the generation of local–global information.

#### D. Model Efficiency and Effectiveness

To fairly compare the model efficiency and effectiveness, we conduct all experiences on the same device with a single NVIDIA GPU 3080. Given the input image of size  $1 \times 3 \times 256 \times 256$ , Table II presents the number of model parameters (Params.), the floating-point operations per second (FLOPs),

the  $F1$ -score and the KC on the three considered CD datasets. We report the inference time of different lightweight methods for inputs with a batch size of 1 on Table III.

The Params. and FLOPs of our MixCDNet are much less than those of the compared transformer-based methods. For example, the CNN-transformer-based method ICIF-Net [29] achieves an  $F1$ -score of 89.66% on the LEVIR-CD dataset with about 23.84M Params. and 25.41G FLOPs. Compared with ICIF-Net, our MixCDNet achieves significantly better detection performance with 0.32M parameters (74.50 times fewer) and 1.59G FLOPs (15.98 times lower). The SNUNet [13] gets  $F1$ -score of 89.31% with about 12.03M Params. and 54.83G FLOPs.

In Fig. 7 the CD performance is intuitively illustrated in terms of  $F1$ -score, number of Params. and computational complexity in terms of FLOPs for several state-of-the-art methods. Compared with four lightweight methods, the proposed MixCDNet has fewer parameters and lower computational complexity than LightCDNet, A2Net, and USSFC-Net. Although TinyCD has slightly fewer Params. and FLOPs, its detection performance is inferior to that of MixCDNet. The complexity comparison results further prove the overall effectiveness of the proposed MixCDNet method.

In terms of model inference efficiency, among lightweight CD methods, the inference time of TinyCD is the lowest and is equal to 34.58 ms to infer a  $3 \times 256 \times 256$  image pair. The inference time of the proposed MixCDNet is 58.80 ms, which is higher than LightCDNet, and lower than A2Net and USSFC-Net. However, the  $F1$ -scores of the proposed MixCDNet are 0.40%/1.77%/3.04% higher than those of TinyCD on the three datasets, respectively. Moreover, compared to LightCDNet, MixCDNet has fewer parameters and lower computational complexity and exceeds the  $F1$ -scores of LightCDNet by 0.73%/1.96%/4.23% on the three datasets, respectively.

#### E. Ablation Studies

To further confirm the contributions of the different components and the proposed MFB and SFSSL modules in our MixCDNet, we perform ablation studies on the CDD dataset.

TABLE II  
COMPARISON OF MODEL EFFICIENCY AND EFFECTIVENESS ON THE THREE TEST SETS. THE SIZE  
OF THE INPUT IMAGE TO THE MODEL IS  $1 \times 3 \times 256 \times 256$

| Method              | Params.(M) | Param. ratio | FLOPs (G) | FLOPs ratio | LEVIR-CD     |              | WHU-CD       |              | CDD          |              |
|---------------------|------------|--------------|-----------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                     |            |              |           |             | F1 (%)       | KC (%)       | F1 (%)       | KC (%)       | F1 (%)       | KC (%)       |
| SNUNet              | 12.03      | 37.59        | 54.83     | 34.48       | 89.31        | 88.75        | 87.85        | 88.33        | 93.82        | 93.00        |
| BIT                 | 3.49       | 11.63        | 10.60     | 6.67        | 90.13        | 89.61        | 88.84        | 88.37        | 94.99        | 94.32        |
| ICIF-Net            | 23.84      | 74.50        | 25.41     | 15.98       | 89.66        | 89.11        | 90.17        | 89.76        | 93.62        | 92.77        |
| LightCDNet          | 1.18       | 8.49         | 2.71      | 1.70        | 90.77        | 89.78        | 91.43        | 90.68        | 92.89        | 91.62        |
| A2Net               | 3.78       | 11.81        | 3.05      | 1.92        | 91.04        | 90.56        | 92.23        | 91.92        | 94.22        | 93.38        |
| USSFC-Net           | 1.52       | 4.75         | 4.86      | 3.06        | 90.27        | 89.69        | 90.78        | 90.39        | 95.16        | 94.46        |
| TinyCD              | 0.29       | 0.91         | 1.54      | 0.97        | 91.10        | 90.63        | 91.62        | 91.27        | 94.08        | 93.21        |
| MixCDNet (proposed) | 0.32       | 1            | 1.59      | 1           | <b>91.50</b> | <b>91.05</b> | <b>93.39</b> | <b>93.12</b> | <b>97.12</b> | <b>96.69</b> |

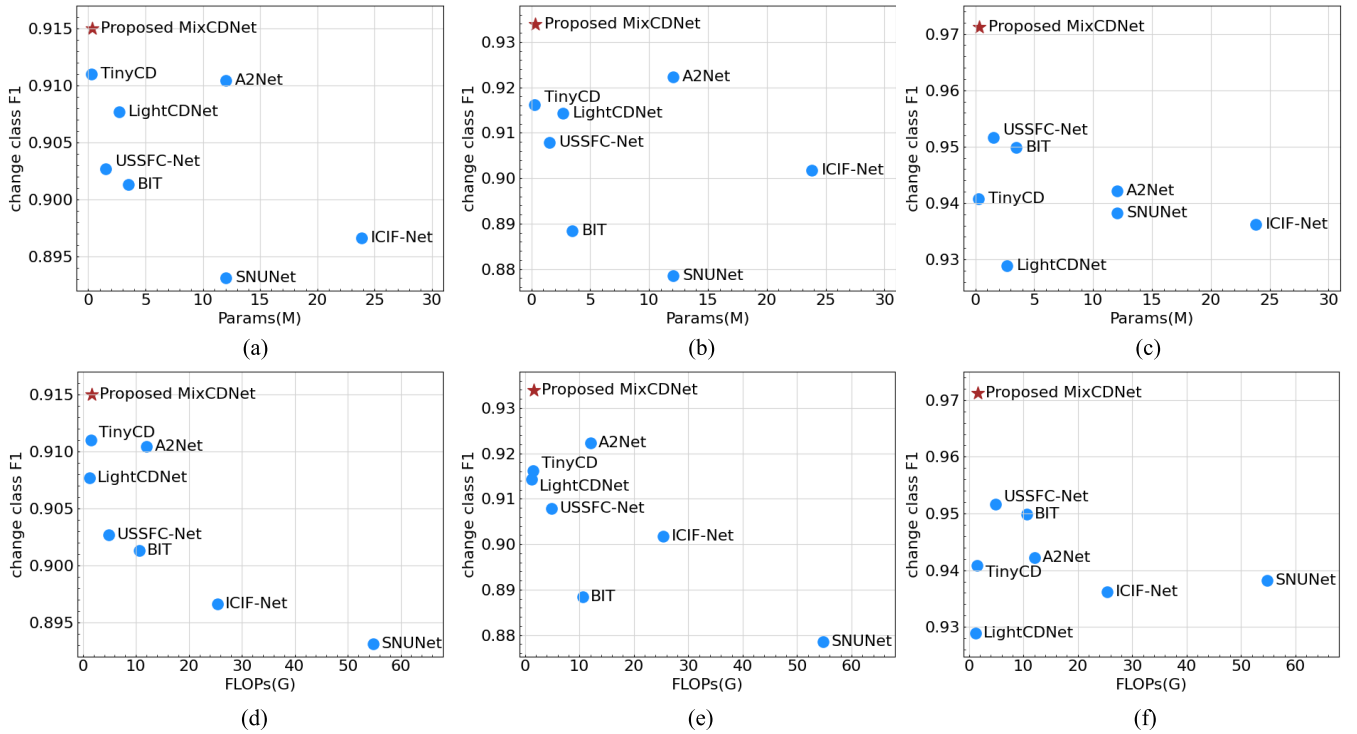


Fig. 7. Model complexity comparison of different methods in terms of Params, FLOPs, and  $F1$ -score on the LEVIR-CD, WHU-CD, and CDD datasets. (a) and (d) LEVIR-CD dataset, (b) and (e) WHU-CD dataset, and (c) and (f) CDD dataset. Our proposed MixCDNet method shows superior performance with fewer parameters and lower computation costs.

TABLE III

INFERENCE EFFICIENCY AND EFFECTIVENESS OF DIFFERENT  
LIGHTWEIGHT METHODS ON THE THREE TEST SETS. THE SIZE  
OF THE INPUT IMAGE TO THE MODEL IS  $1 \times 3 \times 256 \times 256$

| Lightweight<br>Methods | Inference<br>Time (ms) | LEVIR-CD<br>F1 (%) | WHU-CD<br>F1 (%) | CDD<br>F1 (%) |
|------------------------|------------------------|--------------------|------------------|---------------|
| LightCDNet             | 46.20                  | 90.77              | 91.43            | 92.89         |
| A2Net                  | 64.53                  | 91.04              | 92.23            | 94.22         |
| USSFC-Net              | 81.19                  | 90.27              | 90.78            | 95.16         |
| TinyCD                 | 34.58                  | 91.10              | 91.62            | 94.08         |
| MixCDNet(proposed)     | 58.80                  | 91.50              | 93.39            | 97.12         |

1) *Effectiveness of MFB*: The MFB is designed to extract hierarchical LFs and GFs, combining them to enhance attention to changes across various scales and strengthen the complementary information between LFs and GFs. To verify the effectiveness of MFB, we replace the GFB with a LFB

and without bidirectional interactions. This method, denoted as wo-MFB (#02 in Table IV), maintains output channel numbers consistent with those of MFB at each hierarchical level. The results indicate a 3.01% decrease in  $F1$ -score compared to MixCDNet. When we use MFB without bidirectional interactions (#03 in Table IV), the result is still inferior to the proposed MixCDNet. This demonstrates the superiority of MFB within the proposed method.

2) *Effectiveness of SFSSL*: The SFSSL is designed to capture global information with low-computational complexity. It consists of LPU, SSL, and an SF block. From Table IV, one can observe that using SFSSL without LPU (#04 in Table IV) results in worse performance compared to the original SFSSL in MixCDNet. This may be because the LPU alleviates the limitations of transformers, which lack inductive bias like CNNs. For the SFSSL without the SF block (#06

TABLE IV  
ABLATION STUDIES OF THE PROPOSED METHOD ON THE CDD TEST SET. THE SIZE OF THE INPUT IMAGE TO THE MODEL IS  $1 \times 3 \times 256 \times 256$

| No.                                                           | Variants                            | Efficiency |          | F1 (%) | KC (%) |
|---------------------------------------------------------------|-------------------------------------|------------|----------|--------|--------|
|                                                               |                                     | Params.(M) | FLOPs(G) |        |        |
| #1                                                            | MixCDNet                            | 0.32       | 1.59     | 97.12  | 96.69  |
| (a) Mixing Features Block (MFB)                               |                                     |            |          |        |        |
| #2                                                            | w/o MFB (only local features)       | 0.29       | 1.38     | 94.11  | 93.24  |
| #3                                                            | MFB(w/o bi-directional interaction) | 0.31       | 1.55     | 96.58  | 96.08  |
| (b) Skip-Connection and Fusion Separable Self-Attention Layer |                                     |            |          |        |        |
| #4                                                            | SFSSL(w/o LPU)                      | 0.32       | 1.59     | 96.85  | 96.38  |
| #5                                                            | SFSSL(w/o SSL)                      | 0.32       | 2.02     | 96.17  | 95.61  |
| #6                                                            | SFSSL(w/o SF block)                 | 0.31       | 1.56     | 96.69  | 96.21  |
| (c) Different Backbones                                       |                                     |            |          |        |        |
| #7                                                            | ResNet18                            | 16.01      | 34.18    | 92.82  | 91.77  |
| #8                                                            | MobileNetV2                         | 2.25       | 1.21     | 89.56  | 88.05  |

TABLE V  
ABLATION STUDY WITH DIFFERENT NUMBERS OF BLOCKS  $N$  IN LFB AND DIFFERENT NUMBERS OF SSLs  $L$  IN GLB ON THE THREE TEST SETS. THE SIZE OF THE INPUT IMAGE TO THE MODEL IS  $1 \times 3 \times 256 \times 256$

| $N$ | $L$ | Params.(M) | FLOPs(G) | Inference Time (ms) | LEVIR-CD     |              | WHU-CD       |              | CDD          |              |
|-----|-----|------------|----------|---------------------|--------------|--------------|--------------|--------------|--------------|--------------|
|     |     |            |          |                     | F1 (%)       | KC (%)       | F1 (%)       | KC (%)       | F1 (%)       | KC (%)       |
| 2   | 8   | 0.31       | 1.58     | 58.15               | 91.51        | 91.06        | 92.22        | 91.90        | 96.90        | 96.45        |
| 3   | 2   | 0.32       | 1.59     | 39.20               | 91.52        | 91.07        | 92.56        | 92.25        | 96.48        | 95.96        |
|     | 4   | 0.32       | 1.59     | 45.90               | 91.50        | 91.05        | 92.51        | 92.20        | 96.94        | 96.49        |
|     | 8   | 0.32       | 1.59     | 58.80               | 91.50        | 91.05        | <b>93.39</b> | <b>93.12</b> | 97.12        | 96.69        |
|     | 16  | 0.32       | 1.59     | 85.99               | 91.49        | 91.04        | 91.50        | 91.14        | <b>97.22</b> | <b>96.81</b> |
| 4   | 8   | 0.32       | 1.60     | 59.60               | <b>91.56</b> | <b>91.11</b> | 92.37        | 92.05        | 96.93        | 96.47        |

in Table IV), there is a drop in the  $F1$ -score. This indicates that the SF block is important for effectively utilizing features across layers and enhancing the representation capability of features. In the SFSSL without SSL method (#05 in Table IV), we use standard MHA in transformers instead of SSL. The results indicate a decrease in performance compared to MixCDNet, and the computational complexity is higher (2.02 G). Therefore, this indicates the effectiveness of the baseline SFSSL design.

3) *Effectiveness of Different Backbones*: We further study the results obtained using different backbones, i.e., ResNet18 and MobileNetV2. For each backbone, we exclusively utilize all initial blocks up to a spatial resolution downsampling rate of 8. From Table IV, we observe that the EfficientNet-b4 utilized in our MixCDNet outperforms ResNet18 and MobileNetV2 in terms of both performance and computational complexity.

#### F. Parameter Analysis

To analyze the effect of the number of blocks ( $N$ ) in LFB and the number of SSLs ( $L$ ) in GFB, we conducted ablation experiments on the LEVIR-CD, WHU-CD, and CDD datasets. The  $F1$ -score and KC of MixCDNet under different parameter settings on the three datasets are presented in Table V.

First, we fixed the number of blocks ( $N = 3$ ) in LFB and varied the value of  $L$  in GFB. Table V shows no significant improvements in the  $F1$ /KC scores on the LEVIR-CD dataset when increasing the value of the  $L$ . This indicates that the global relationships within the bitemporal images in this dataset can be well-learned by a relatively simple GFB structure. One possible reason is that the LEVIR-CD dataset only contains changes in buildings and has limited scene complexity. For the WHU-CD dataset, the  $F1$ -score is highest when  $L$  is set to 8. When  $L$  is set to 16, the performance is the lowest, possibly indicating overfitting due to the relatively small size of the dataset. Regarding the CDD dataset, we note a gradual performance improvement when increasing the value of  $L$ , but the magnitude of performance improvement becomes increasingly smaller. Therefore, we believe that setting  $L$  to 8 strikes a good trade-off between efficiency and precision. One can observe that increasing the value of  $L$  does not lead to a significant increase in the number of Params. and FLOPs of the model, while there is an obvious increase in inference time. To further investigate the effect of the number of blocks ( $N$ ) in LFB, we fixed  $L$  at 8 and varied the values of  $N$ . One can see that setting  $N$  to 3 results in higher performance on the three datasets. In general, increasing  $N$  results in a corresponding increase in the number of Params. and FLOPs of the model.

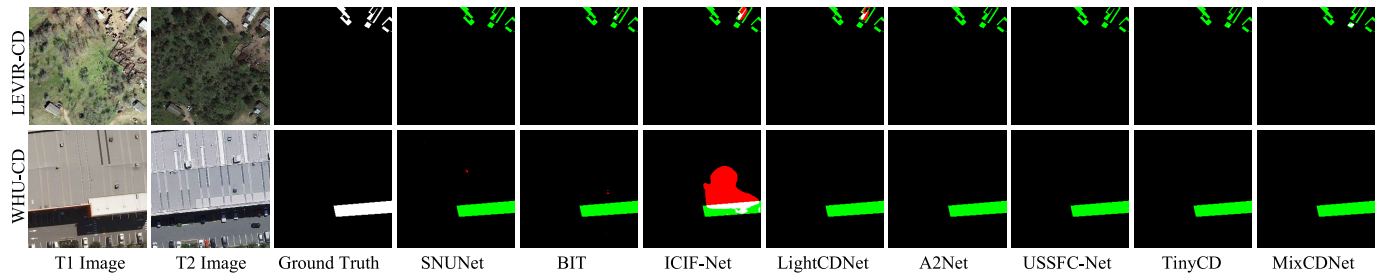


Fig. 8. Visual results of failure cases of different methods on LEVIR-CD and WHU-CD test set. For a better view, white, black, red, and green represent TP, TN, FP, and FN, respectively.

### G. Discussions About Failure Cases

The proposed method MixCDNet has demonstrated excellent performance and can detect changed areas in the most challenging scenarios, as discussed in Section IV-C. However, as shown in Fig. 8, in some cases we also observed some incorrect detection results. In the first row, the changed area involves both demolition and expansion of the buildings, resulting in only partial changes to the main structure. In the second row, the changed area corresponds to the demolished building, but the post-demolition region is relatively small in size compared to the unchanged original building. This change failed to be accurately detected by all the mentioned comparative methods and also the proposed MixCDNet method. In cases where only partial changes occur in the main structure of a building, more refined CD techniques are required. Furthermore, we need to continue exploring how to more accurately detect the edges of buildings to assist in the building's CD task.

### V. CONCLUSION

In this article, we have proposed MixCDNet, which is a super lightweight hybrid transformer-based CD network with only 0.32M Params. and 1.59G FLOPs. The proposed CD method extracts both LFs and GFs by introducing MFB and SFSSL. Specifically, the MFB extracts local and global hierarchical features and mixes them through bidirectional interactions. The SFSSL effectively captures global information using lightweight SSLs with linear complexity. Extensive experiments have validated the ability of the proposed method to effectively mitigate errors caused by varying illumination intensities and interference from similar appearance ground targets in bitemporal images, as well as its superior performance in detecting dense and small-scale changes. Results on three HRRS CD datasets have shown that MixCDNet exhibits good generalization capability and outperforms comparison methods in terms of overall detection performance and model complexity.

In future research endeavors, we aim to conduct further exploration to enhance detection capabilities in complex backgrounds, particularly addressing scenarios where buildings undergo small and partial changes.

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